00-D-565a
December 12, 1967
SUPERSEDING
Fed. Spec. 00-D-565
January 4, 1956

FEDERAL SPECIFICATION

DOGS, LATHE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers standard and parallel, straight and bent tail lathe dogs.

1.1.1 Federal specification coverage. This specification does not include all types, classes, and styles of lathe dogs but is intended to cover only those types, classes, and styles generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Lathe dogs shall be furnished in the following types, classes, and styles, as specified (see 6.2):

Type I. Standard lathe dogs.
  Class A. Straight tail.
    Style 1. Single screw.
    Style 2. Double screw.
  Class B. Bent tail.
    Style 1. Single screw.
    Style 2. Double screw.
Type II. Parallel (clamp) lathe dogs.
  Class A. Straight tail.
  Class B. Bent tail.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-562 - Box, Corrugated, Fiberboard, High Compression Strength, Water-Resistant, Double-Wall.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

United States of America Standards Institute (USASI) Standard:

B18.3-1961 - Socket Cap, Shoulder and Set Screws.

(Application for copies should be addressed to the United States of America Standards Institute (USASI), 10 East 40th St., N. Y., N. Y. 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of lathe dogs otherwise conforming to this specification.

3.2 Material. Material used in fabrication of lathe dogs shall be new and of a quality necessary to produce lathe dogs to meet the requirements specified herein. Bodies of standard lathe dogs and clamping bars of parallel dogs shall be of deep forged steel.

3.3 Construction. The lathe dog shall be free of any characteristics or defects that will prevent the lathe dog from passing the examinations and tests specified in section 4

3.3.1 Forgings. Forgings shall be free from defects and scale. In no event shall such processes as welding, peening, plugging, or filling with cold solders or metallic pastes be used.

3.4 Surface finish. Exposed surfaces shall be smooth and edges shall be rounded or beveled. Surfaces shall be coated or finished according to the manufacturer's standard practice.

3.5 Screws. Screws shall have American National coarse series or American National 8-pitch series, right-hand threads, class 2A or class 3A fit, conforming to H28.

3.6 Design. The design shall be such as to perform the functions of the types and classes specified. All parts shall be new, designed, and constructed to the requirements specified herein.

3.7 Type I, standard lathe dogs. Type I standard lathe dogs shall be suitable for driving cylindrical work. They shall be of the safety dog design, incorporating recessed head socket set screws. Type I lathe dogs shall be furnished in class A or class B, as specified (see 6.2).

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3.7.1 Class A, straight tail. Class A lathe dogs shall be furnished in style 1 or style 2, as specified (see 6.2).

3.7.1.1 Style 1, single screw. Style 1 lathe dogs shall conform to table I for the size specified (see 6.2), and shall be similar to figure 1. They shall be approximately elliptical in shape with a hub or boss at the top. The hub or boss shall be drilled and tapped to receive a screw. The length of the threaded hub shall be not less than twice the diameter of the set screw. The bottom end shall form a tail parallel with the body.

TABLE I. Size, capacity, and dimensions of type I, class A, standard, straight tail lathe dog

Size	Bar capacity (A)		Minimum screw diameter (B)	Thickness through hole (C) (plus or minus 20 percent)	Length overall without screw (D) (plus or minus 1 inch)
	At least	Maximum tolerance (plus)			
	Inches	Inch	Inch	Inches	Inches
1	3/8	1/16	5/16	9/16	3-1/2
2	1/2	1/16	3/8	5/8	3-5/8
3	3/4	1/16	7/16	3/4	4-1/8
4	1	1/16	1/2	7/8	5
5	1-1/4	1/8	9/16	1	5-3/4
6	1-1/2	1/8	9/16	1-1/16	6-1/2
7	1-3/4	1/8	5/8	1-1/8	7-1/4
8	2	3/16	5/8	1-1/8	7-3/4
9	2-1/2	3/16	3/4	1-3/8	9-1/16
10	3	3/16	3/4	1-1/4	10-1/4
11	3-1/2	1/4	7/8	1-5/8	11-1/4
12	4	1/4	7/8	1-1/2	12-3/8
13	5	3/8	1	1-7/8	14-1/4
14	6	3/8	1	3	15-1/2

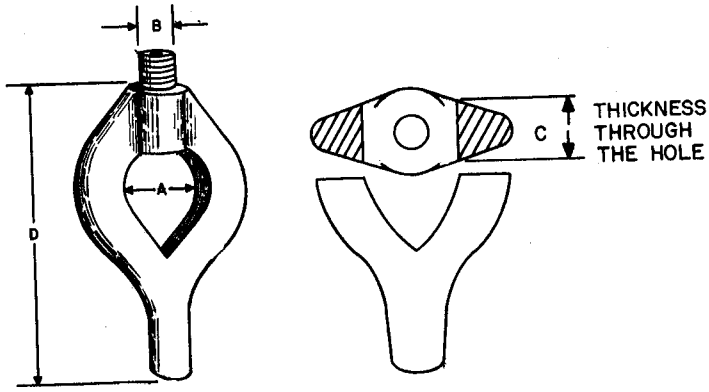


FIGURE 1. Type I, class A, style 1, standard, straight tail, single screw lathe dog.

3.7.1.1.1 Set screw. A headless hexagon socket type set screw and a matching Allen wrench (hex key) shall be supplied with each dog. The screw shall be of alloy steel hardened and tempered to Rockwell C 45 to 53 and of such a length that the head of the screw is flush with the top of the tapped hole when the dog is carrying its capacity size. Screws shall have a full or half dog point conforming to standards for socket set screws in accordance with B18.3-1961.

3.7.1.2 Style 2, double screw. The body and screw design of class A, style 2, lathe dogs shall conform to the requirements of style 1 (see 3.7.1.1) except that the body shall have a double hub or boss at the top, both threaded to receive a screw, and two screws shall be supplied with each lathe dog. The center line of the tapped holes, if extended, shall cross approximately at the center of a capacity size stock inserted in the dog. Style 2 dogs shall conform to table I for the size specified (see 6.2), and shall be similar to figure 2.

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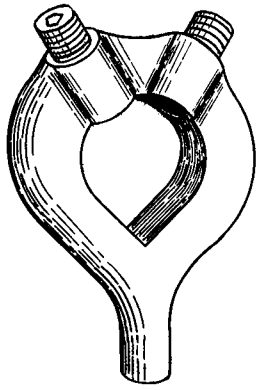


FIGURE 2. Type I, class A, style 2, standard, straight tail, double screw lathe dog.

3.7.2 Class B, bent tail. Type I, class B lathe dogs shall be style 1 or style 2, as specified (see 6.2).

3.7.2.1 Style 1, single screw. The body and screw design of class B, style 1 lathe dogs shall conform to the requirements for class A, style 1 (see 3.7.1.1), except that the included angle of the tail and the face of the dog shall be approximately 97°. Class B, style 1 lathe dogs shall conform to table II for the size specified (see 6.2), and shall be similar to figure 3.

TABLE II. Sizes, capacities, and dimensions of type I, class B, standard, bent tail lathe dog, styles 1 and 2

Size	Bar capacity (A)		Minimum screw diameter (B)	Thickness through hole 1/ (plus or minus 20 percent)	Length overall (D) without screw (plus or minus 1 inch)	Length of tail (E) (plus or minus 1 inch)
	At least	Maximum tolerance (plus)				
	Inches	Inch	Inches	Inches	Inches	Inches
1	3/8	1/16	3/16	9/16	3-3/16	2-1/2
2	1/2	1/16	3/16	5/8	3-11/16	2-3/4
3	3/4	1/16	7/16	3/4	4-1/16	2-3/4
4	1	1/16	7/16	7/8	4-1/2	2-7/8
5	1-1/4	1/8	9/16	1	5-1/32	3-1/8
6	1-1/2	1/8	9/16	1-1/16	5-5/8	3-1/4
7	1-3/4	1/8	5/8	1-1/8	6-1/4	3-3/8
8	2	3/16	5/8	1-1/8	6-9/16	3-5/8
9	2-1/2	3/16	3/4	1-3/8	7-5/16	3-7/8
10	3	3/16	3/4	1-1/2	7-15/16	4-1/8
11	3-1/2	1/4	7/8	1-5/8	9	4-1/2
12	4	1/4	7/8	1-3/4	9-11/16	4-7/8
13	5	3/8	1	1-7/8	10-7/8	5-1/2
14	6	3/8	1	2	13	5-1/2
15	7	3/8	1-1/8	2-1/8	17	7
16	8	3/8	1-1/8	2-1/4	19	7

1/ See figure 1.

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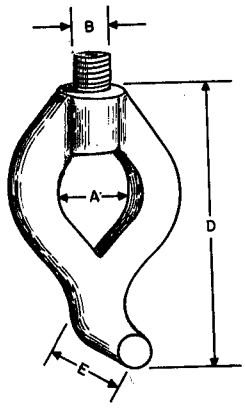


FIGURE 3. Type I, class B, style 1, standard, bent tail, single screw lathe dog.

3.7.2.2 Style 2, double screw. The body and screw design of class B, style 2, lathe dogs shall conform to the requirements for class A, style 2 (see 3.7.1.2) except that the included angle of the tail and the face of the dog shall be approximately 97°. Class B, style 2, lathe dogs shall conform to table II for the size specified (see 6.2), and shall be similar to figure 4.

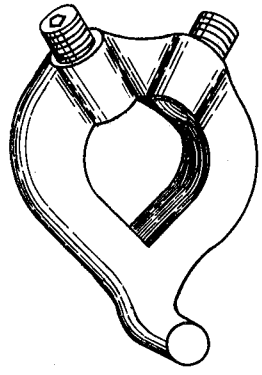


FIGURE 4. Type I, class B, style 2, standard, bent tail, double screw lathe dog.

3.8 Type II, parallel (clamp) lathe dogs. Type II parallel lathe dogs shall be suitable for driving work with flat sides. Type II lathe dogs shall be class A or class B, as specified (see 6.2).

3.8.1 Class A, straight tail. Type II, class A lathe dogs shall consist of two parallel clamping bars, together with two adjusting screws. The clamping bars shall be steel forgings heat treated and machined. The matching faces of the bars shall be machine finished. Small matching "V" slots shall be machined on the face of each bar, midway between the screws, and at right angles to the face of the bar. The top bar shall have a hole in each end to receive the screws with sufficient clearance to allow for considerable tilt without bending the screw. The bottom bar shall have two threaded holes to receive the screws, spaced to match the holes in the top bar. One end of the bottom bar shall extend to form a tail parallel with the face of the dog. The screws shall have a square head and a shoulder or washer under the head to prevent binding. The screws shall conform to the requirements of H28. Type II, class A, lathe dogs shall conform to table III for the size specified (see 6.2), and shall be similar to figure 5.

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TABLE III. Sizes and capacities of type II, classes A and B, parallel straight and bent tail lathe dogs

Nominal size	Clearance between bolts (B) (plus 1/4, minus 0 inch)	Width of bar (C) (plus 1/4, minus 0 inch)	Length of tail, dimension (A) (plus 1/2, minus 0 inch)	Screw length under head (D) (plus or minus 1/2 inch)	Screw diameter (E) (plus 1/8, minus 0 inch)
Inches	Inches	Inches	Inches	Inches	Inch
1-1/2	1-13/16	9/16	2-7/32	2-3/4	5/16
1-7/8	2-5/16	5/8	2-1/2	3-5/16	3/8
2-1/2	2-13/16	3/4	3	4-9/32	7/16
3	3-5/8	7/8	3-1/4	5-13/32	1/2
4	4	1-1/2	3-3/4	7	3/4

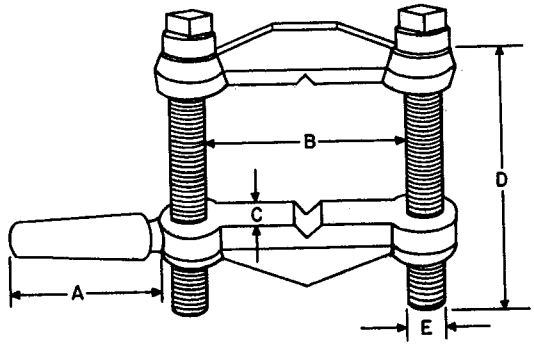


FIGURE 5. Type II, class A, parallel, straight tail lathe dog.

3.8.2 Class B, bent tail. The body and screw design of class B lathe dogs shall conform to the requirements for type II, class A (see 3.8.1), except that the included angle of the tail and the face of the dog shall be approximately 97°. Type II, class B lathe dogs shall conform to table III for the size specified (see 6.2), and shall be similar to either figure 6 or 7.

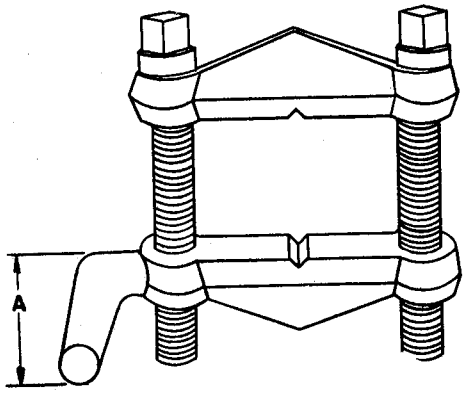


FIGURE 6. Type II, class B, parallel, bent tail lathe dog.

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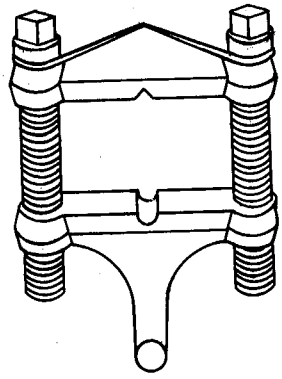


FIGURE 7. Type II, class B, parallel, bent tail lathe dog.

3.9 Identification. Each lathe dog shall be permanently and legibly marked with the manufacturer's name or trademark and size number.

3.10 Workmanship. Workmanship of lathe dogs shall be of a quality prevailing among manufacturers normally producing lathe dogs of the types specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. All lathe dogs of the same type, class, style, and size presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.2 Sampling for quality conformance inspection. A random sample of lathe dogs shall be selected from each lot in accordance with MIL-STD-105 at inspection level I.

4.3 Quality conformance inspection. Quality conformance inspection shall be performed on each sample lathe dog selected in accordance with 4.2.2. Quality conformance inspection shall consist of the following:

(a) Visual and dimensional examination (see 4.3.1).
(b) Tests (see 4.3.2).
(c) Examination of preservation, packaging, packing, and marking for shipment and storage (see 4.3.3).

4.3.1 Visual and dimensional examination. Each lathe dog selected in accordance with 4.2.2 shall be examined visually and dimensionally to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.3.2 Tests. After visual and dimensional examination, each type I lathe dog selected in accordance with 4.2.2 shall have the set screws removed and the set screw hardness determined for compliance with the hardness requirements of 3.7.1.1.1 and B18.3-1961. Hardness readings shall be made on the dog point end. Failure to comply with the hardness range specified shall be cause for rejection.

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4.3.3 Inspection of preparation for delivery. An inspection shall be made to determine that the cleaning, preservation, packing and marking comply with the requirements of this specification. Defects shall be scored in accordance with table IV. Sampling shall be in accordance with MIL-STD-105, inspection level S-2, with an acceptable quality level (AQL) of 4.0 defects per hundred units. The lot size shall be the number of shipping containers in the end item inspection lot.

TABLE IV. Classification of preparation for delivery defects

Examine	Defects
Cleaning and preservation	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Not as specified. Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container, loose strapping or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and preservation. Cleaning and preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A. All external metal surfaces shall be cleaned in accordance with MIL-P-116, method C-1, and coated with P-1 preservative in accordance with MIL-P-116.

5.1.2 Level C. Each lathe dog shall be cleaned and preserved in accordance with the supplier's normal commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Dog lathes shall be packed in nailed wood, cleated-plywood or fiberboard boxes conforming to PPP-B-621, class 2; PPP-B-601, overseas type; PPP-B-640, class 1; PPP-B-636, class weather resistant; or PPP-B-562.

5.2.2 Level B. Dog lathes shall be packed in a nailed wood, cleated-plywood or fiberboard boxes conforming to PPP-B-621, class 1; PPP-B-601, domestic type; PPP-B-640, class 2; or PPP-B-636, class domestic. Gross weight of boxes shall not exceed 65 pounds.

5.2.3 Level C. Dog lathes shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order (see 6.2), the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order (see 6.2), the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Lathe dogs covered by this specification are used on lathes and cylindrical grinders for driving a workpiece held between centers. Type I dogs are used for driving cylindrical pieces. Type II dogs are used for work with flat sides.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style (see 1.2.1).
(c) Size (see applicable table).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) Marking requirements (see 5.3).

MILITARY CUSTODIANS:
Army - WC
Navy - SH
Air Force - 84

Review activities:

Army - WC
Navy - SH
Air Force - 84

User activities:

Navy - YD, MC

Preparing activity:

Navy - SH

CIVIL AGENCY INTEREST:

GSA-FSS

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